Snowflake Enablement: Cortex Code

Haley Massa, Sr. Applied Field Engineer AI / ML

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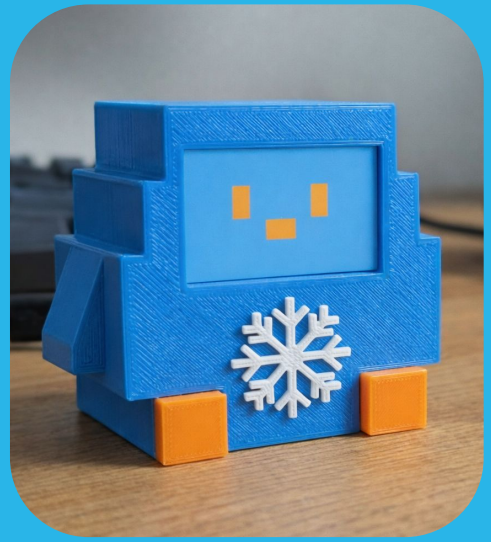

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Native AI & Agents Inside Snowflake

SNOWFLAKE CORTEX AI

AGENTIC BUSINESS INSIGHT

- Agent Apps **Snowflake Intelligence**
- Agent APIs **Cortex Agents**

STATE OF THE ART RETRIEVAL

- Unstructured Data Retrieval **Cortex Search**
- Structured Data Retrieval **Cortex Analyst**

SCALABLE AI PROCESSING

- Multimodal AI-powered SQL **Cortex Functions**
- Document Processing **Parse, Embed**

MODELS

OpenAI
ANTHROP\C
Gemini
Meta
MISTRAL AI_
deepseek
snowflake

GOVERNANCE

- RBAC
- Guardrails
- Evaluations
- Monitoring
- AI Gateway

DATA STR DOC AUDIO IMAGE

Coding / Building
Cortex Code

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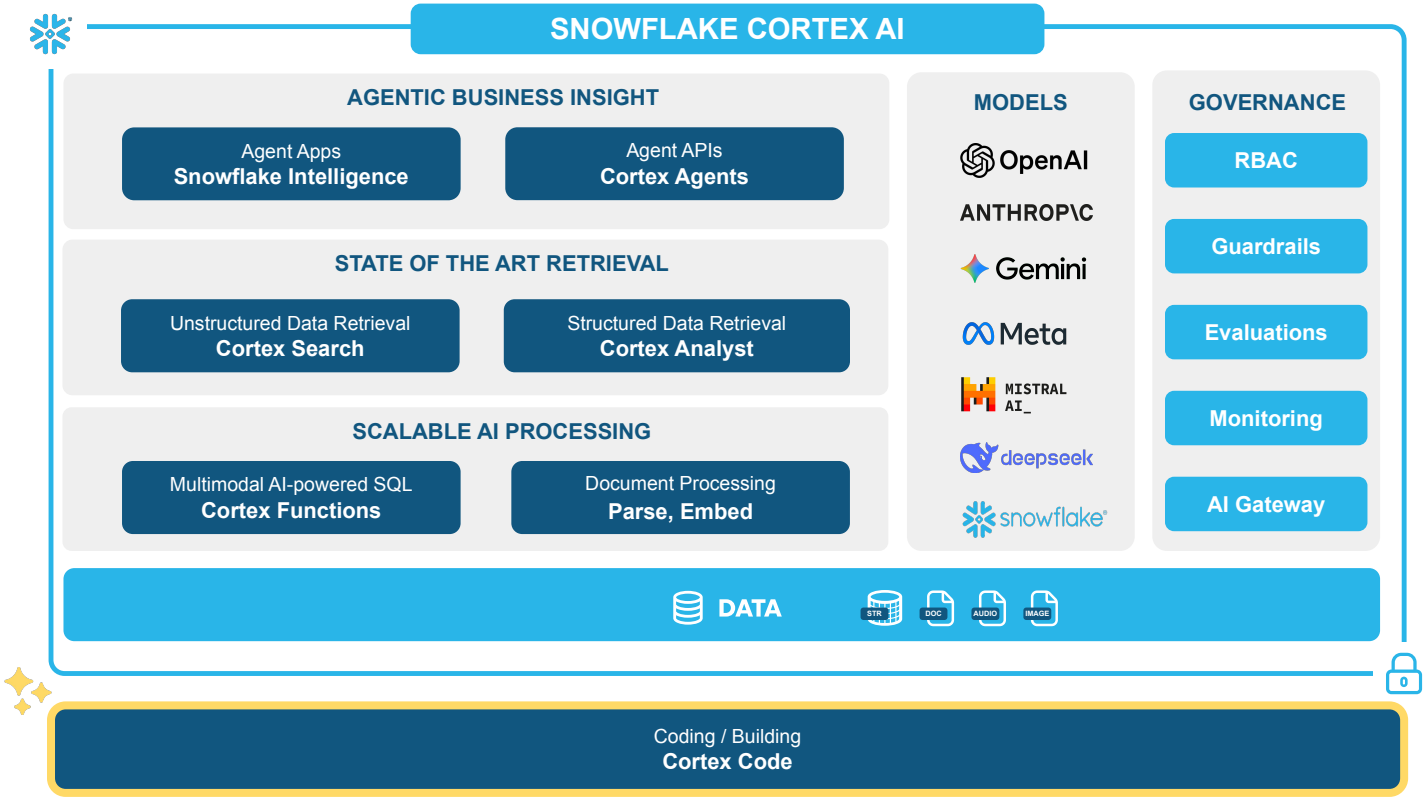
Native AI & Agents Inside Snowflake

Cortex Code

AI coding agent that generates and iterates on Snowflake-native code (SQL, Python, Streamlit, etc.) grounded in your data and environment.



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The Right Tool for Data Teams

An AI Coding Agent built to get you to production faster

UNDERSTANDS SNOWFLAKE

A built-in expert on Snowflake, it has up-to-the-minute knowledge of your specific data catalog: databases, schemas, semantic models and more.

INTELLIGENCE

CONTEXT AWARE

The agent knows where you are in the UI, what data you're working with, and retains your conversational history as you navigate.

RELEVANCE

AUTOMATES COMPLEX TASKS

Integrates natively with your stack. dbt, Git, SQL, Python, and Streamlit. No changes to how you work.

INTEGRATION

SECURE BY DESIGN

Cortex Code uses Snowflake RBAC, ensuring it only sees and acts on the data and objects you have access to

GOVERNANCE

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Cortex Code: AI Coding Agent for Data Teams

THE CURRENT STATE

Traditional Dev Friction

FRAGMENTED TOOLCHAIN
Workflows spanning dbt, Airflow, SQL, and Notebooks → constant context switching.

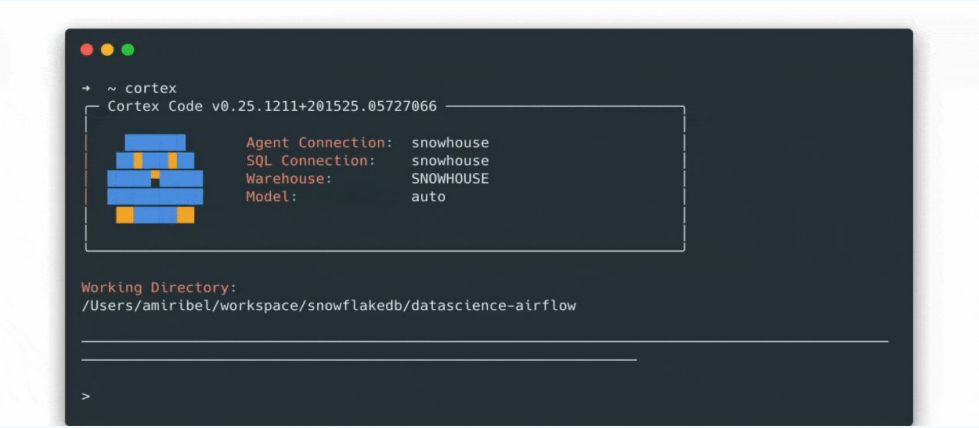
SLOW ITERATION
Manual refactoring and tribal knowledge slow debugging and deployments.

GOVERNANCE FRICTION
Access controls and policies bottleneck rapid prototyping.

THE SOLUTION

AI-Powered Development

From natural language to production-ready code across your entire data stack.
CLI & in Snowsight.



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Accelerate Your Data and AI Workflows

CORTEX CODE

DATA DISCOVERY
Search the catalog for data with semantic understanding

ADMIN TASK
Manage users, permissions, costs and governance

DATA ENGINEERING
Build & optimize pipelines with dbt and AirFlow

AI & AGENTS
Build, evaluate, optimize, debug agents and semantic views

ML & DATA SCIENCE
Auto-generate ML pipelines and feature engineering

ADVANCED ANALYTICS
Complex Analysis with Python and SQL generation

BI & APPS
Streamlit apps, dashboards & data applications

CODE MIGRATION & REFACTORING
Refactor and convert code from other platforms

UNDERSTANDS SNOWFLAKE & DATA | **AUTOMATES COMPLEX TASKS** | **CONTEXT AWARE CLI & UI** | **SECURE BY DESIGN**

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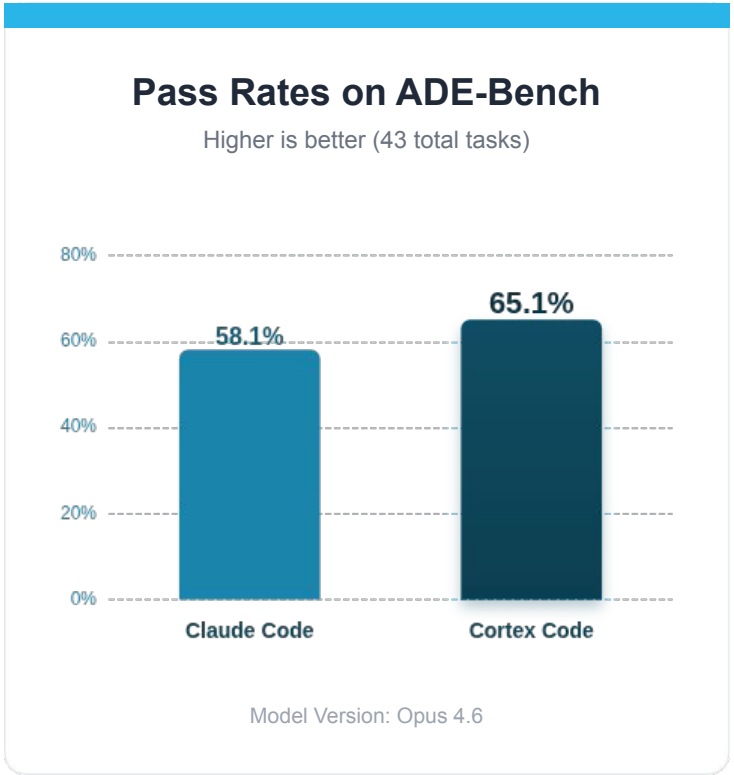
Cortex Code Is the World's Best Data Coding Agent



BENCHMARK RESULTS

Cortex Code achieves comparable rates but demonstrates superior execution efficiency, with a **15.7% higher efficiency score** compared to Claude Code.

TOOL CALLS	AVG STEPS	OPERATIONS
854 ↓ 48%	**19.9** ↓ ~50%	File Reads: 2x Fewer
vs 1,657 (Claude)	vs 38.5 (Claude)	Bash Cmds: 4x Fewer

Why Cortex Code is More Efficient

TARGETED VS. EXHAUSTED

Cortex performs targeted exploration rather than exhaustive scanning, significantly reducing IO operations.

SQL-NATIVE APPROACH

Utilizes **snowflake_sql_execute** for native execution, avoiding heavy reliance on bash tools common in Claude's workflow.

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* Analytics and Data Engineering Benchmark by DBT

Snow-on-Snow: 10x Faster Root Cause Analysis for CX

10X

Faster Resolution

BEFORE
1 Hour to Days

AFTER
15 mins to 2 hrs

JIRA QUALITY
"Help us debug" → "Confirm this fix"

ENG EFFORT
Reduced from days to minutes

Context | Getting the right context together

Semantic Views
GET_ENVOY_LOGS, GET_JOBS for correct query results

Tool Integration
MCP connections to 10+ systems (Snowflake, SFDC, JIRA, GitHub, Glean, Confluence, etc.)

Intelligence | Data agent with encoded investigation patterns

Skills & Runbook
Reusable prompts encoding investigation patterns ("How do I debug throttling?")

Planning & Reasoning
Dynamic investigation paths & anomaly detection

Interface | Surfaced to Engineers in Cortex Code

Profiles
Packaged configurations (skills + tools + settings) stored centrally

No Setup Required
Any engineer gets the same toolset instantly available

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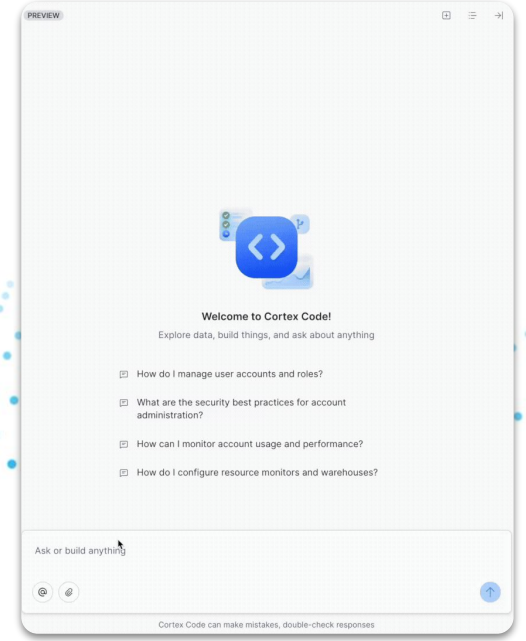
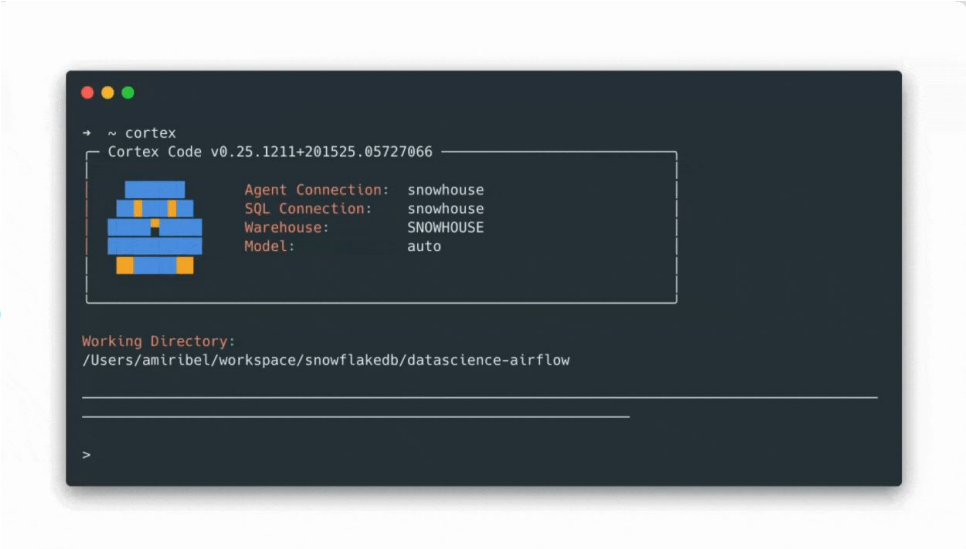
Where to Access Cortex Code





Snowsight

CLI

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CREATING COCO CLI CONNECTION

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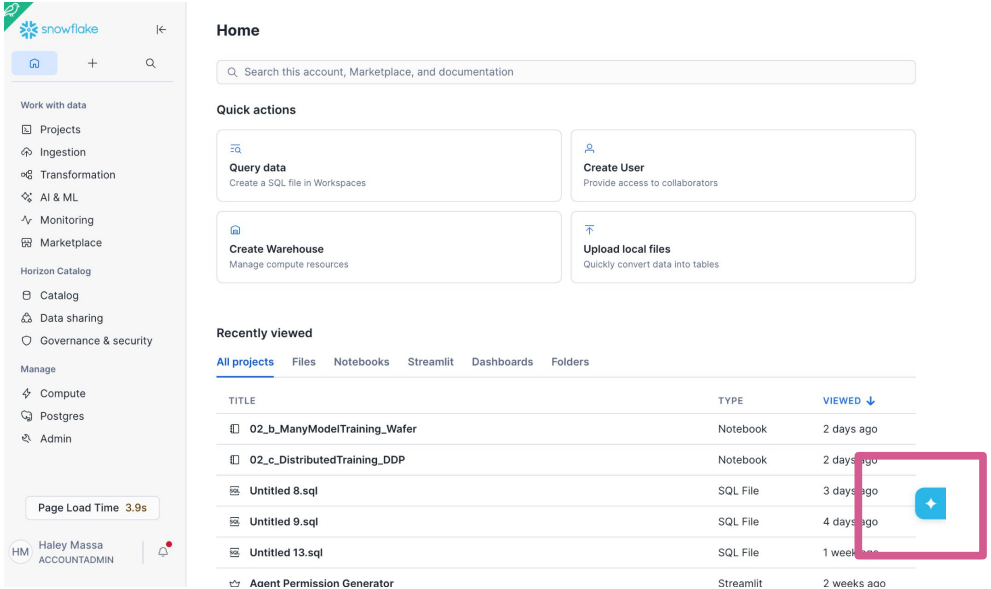
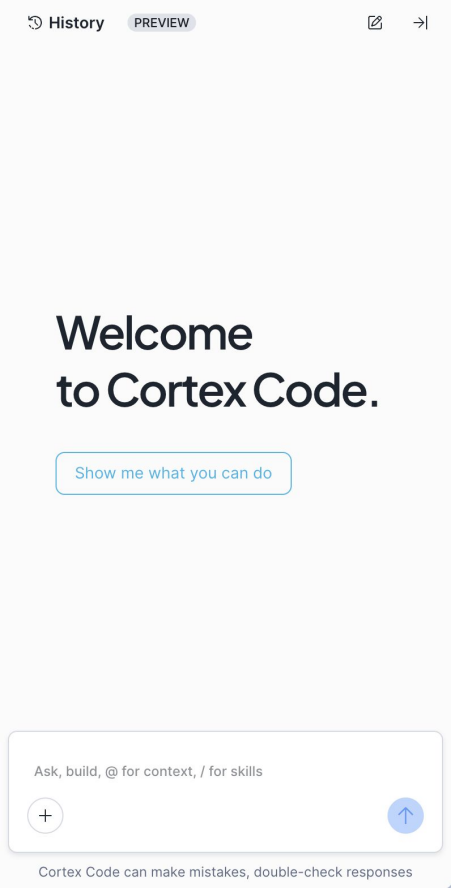
Validate Cortex Code in Snowsight





Access Cortex Code via Snowsight by clicking on the blue star in the bottom right corner in the UI

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Installing the CoCo CLI on Mac

INSTALL VIA TERMINAL

```
curl -LsS https://ai.snowflake.com/static/cc-scripts/install.sh | sh
```

Architectures

- x64 – Supported.
- ARM64 – Supported

VERIFY INSTALL

```
cortex --version
```

During Installation

- When prompted to add cortex to your PATH, respond with y.
- If the cortex command is not recognized after installation:
- Close and reopen your terminal, or
- Restart your machine to ensure PATH changes take effect.

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Installing the CoCo CLI on Mac

CONFIGURE SNOWFLAKE CONNECTIONS

- Cortex Code uses the same connection files as SnowCLI, under your home directory. On Macs that's typically:
- ~/.snowflake/config.toml (or connections.toml if you already use that).
- Minimal example (adapt to your accounts/roles):

```
default_connection_name = "DEMO"  # NAME FOR DEFAULT SNOWFLAKE CONNECTION
[connections.DEMO]  # NAME FOR SNOWFLAKE CONNECTION (SAME AS ABOVE IF DEFAULT)
account   = "<YOUR_DEMO_ACCOUNT>"      # e.g. ACCOUNT IDENTIFIER
user      = "<YOUR_DEMO_USERNAME>"     # e.g. AUTHENTICATION VARIABLE
password  = "<YOUR_DEMO_PAT>"
role      = "<YOUR_DEMO_ROLE>"         # e.g. NEED A ROLE FOR CONNECTION
warehouse = "<YOUR_DEMO_WAREHOUSE>"
```

LAUNCH CORTEX CODE CLI USING YOUR SQL CONNECTION

```
cortex -c DEMO # connect to connection name
```

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Installing the CoCo CLI on Windows

INSTALL VIA POWERSHELL

```
irm https://ai.snowflake.com/static/cc-scripts/install.ps1 | iex
```

Architectures

- x64 – Supported.
- ARM64 – Support is in progress. Current builds may rely on experimental runtime support and are not yet recommended for production use.

VERIFY INSTALL

```
cortex --version
```

During Installation

- When prompted to add cortex to your PATH, respond with y.
- If the cortex command is not recognized after installation:
  - Close and reopen your terminal, or
  - Restart your machine to ensure PATH changes take effect.

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Installing the CoCo CLI on Windows

CONFIGURE SNOWFLAKE CONNECTIONS

- Cortex Code uses the same connection files as SnowCLI, under your home directory. On Windows that's typically:
- %USERPROFILE%.snowflake\config.toml (or connections.toml if you already use that).
- Minimal example (adapt to your accounts/roles):

```
default_connection_name = "DEMO"  # NAME FOR DEFAULT SNOWFLAKE CONNECTION
[connections.DEMO]  # NAME FOR SNOWFLAKE CONNECTION (SAME AS ABOVE IF DEFAULT)
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user      = "<YOUR_DEMO_USERNAME>"     # e.g. AUTHENTICATION VARIABLE
password  = "<YOUR_DEMO_PAT>"
role      = "<YOUR_DEMO_ROLE>"         # e.g. NEED A ROLE FOR CONNECTION
warehouse = "<YOUR_DEMO_WAREHOUSE>"
```

LAUNCH CORTEX CODE CLI USING YOUR SQL CONNECTION

```
cortex -c DEMO # connect to connection name
```

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Creating a Snowflake Connections Config

ON MAC

Run this code in your terminal to create the connections config file

```
mkdir -p ~/.snowflake
touch ~/.snowflake/config.toml
chmod 600 ~/.snowflake/config.toml
```

Then run this command to open the file to edit and add in config

```
open -e ~/.snowflake/config.toml
```

ON WINDOWS

Run this code in powershell to create the connection config file

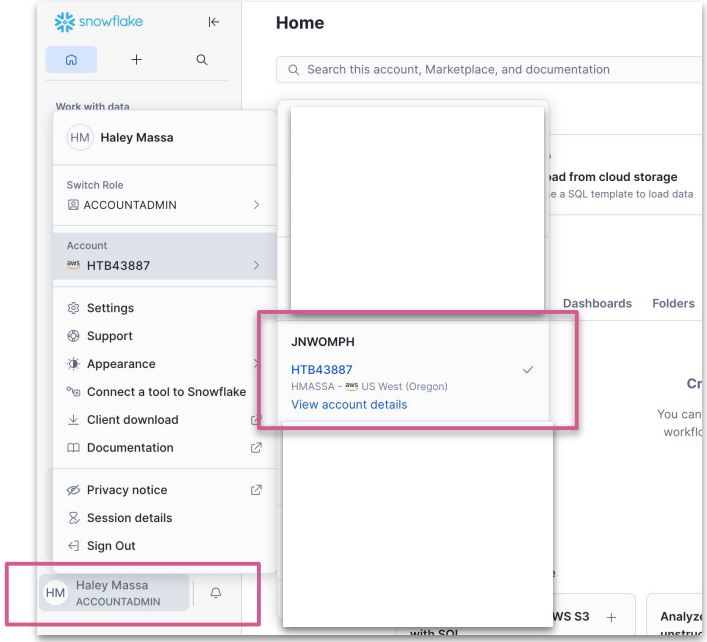
```
mkdir $env:USERPROFILE\.snowflake -Force
New-Item -ItemType File -Path $env:USERPROFILE\.snowflake\config.toml
-Force
```

Then run this command to open the file to edit and add in config

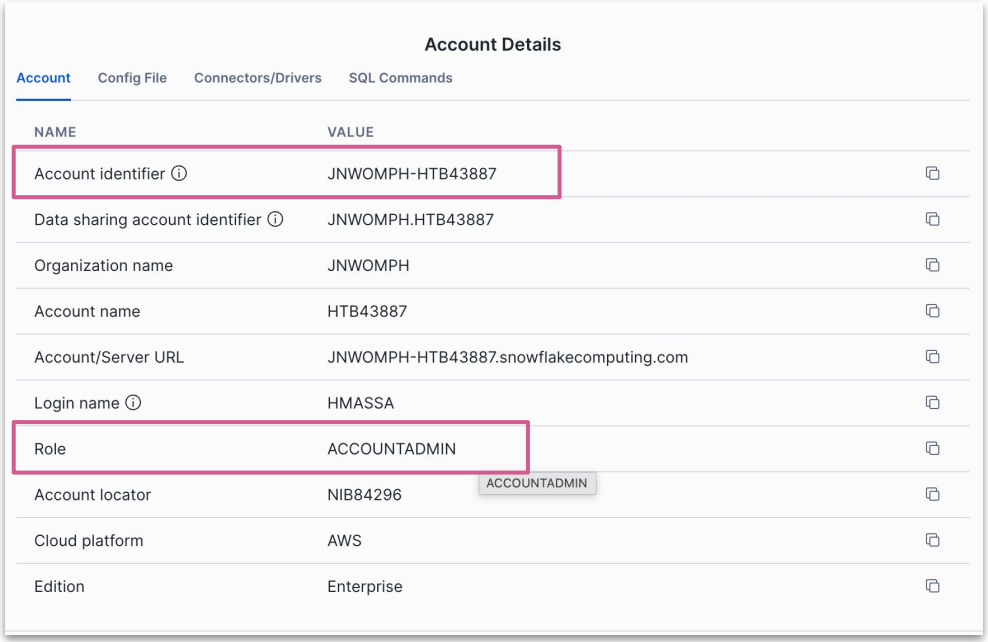
```
notepad $env:USERPROFILE\.snowflake\config.toml
```

Filling out the Default Connection

LOG IN AND FIND SETTING INFORMATION



CLICK ON CONNECTION AND FIND DEFAULTS



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Ways to Authenticate Connections

CREATE AND AUTHENTICATE A CONNECTION TO SNOWFLAKE

There are many ways to create a connection to Snowflake. These are included into your connection configuration as credentials to identify you as logging into your connection. Here are some examples of environment variables used in Snowflake credentials:

- `SNOWFLAKE_ACCOUNT`
- `SNOWFLAKE_USER`
- `SNOWFLAKE_PASSWORD`
- `SNOWFLAKE_TOKEN`
- `SNOWFLAKE_TOKEN_FILE_PATH`
- `SNOWFLAKE_OAUTH_CLIENT_ID`
- `. . .`

Find more examples and walk through creating your connection parameter in the [Manage Snowflake connections guide](#).

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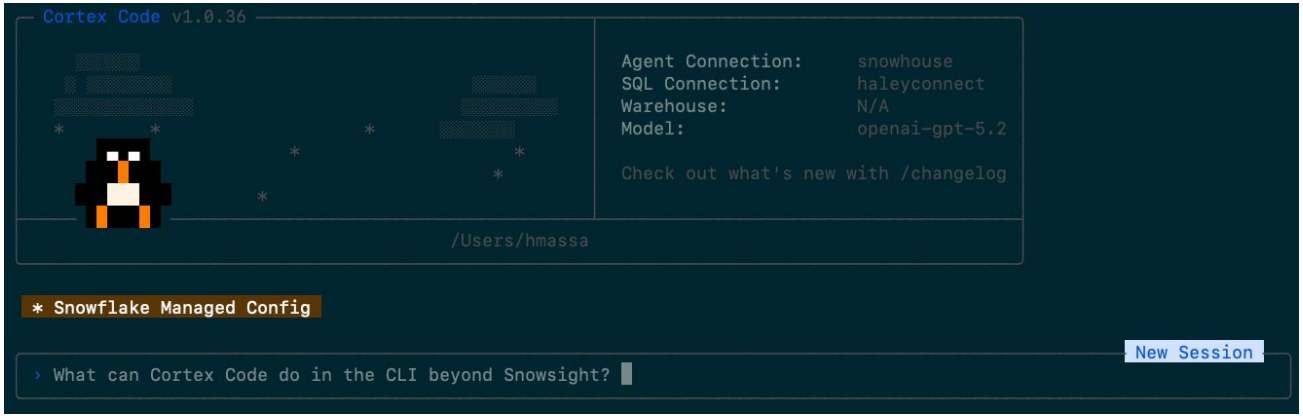
Launching Cortex Code

LAUNCH CORTEX CODE CLI USING YOUR SQL CONNECTION

```
cortex -c DEMO # connect to connection name
```

- After connecting to cortex code (using the connect -c command and your connection name (in this example, DEMO), you will see the Cortex Code CLI interface
- From here is where we will cover more advanced usages of Cortex Code in our session, but we can test the connection first by asking a simple question:



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CoCo as an MCP Server in VS Code

CONFIGURE COCO CLI AS AN MCP SERVER IN VS CODE

- VS Code's GitHub Copilot supports MCP servers natively.
- You register Cortex Code CLI as a stdio MCP server so Copilot can invoke its Snowflake tools.
- Create .vscode/mcp.json in your project root:

```
{"servers": {
   "cortex-code": {
     "type": "stdio",
     "command": "cortex",
     "args": ["mcp"]
   }}}
```

VERIFY THE CONNECTION

- Open the Command Palette (Ctrl+Shift+P) -> "MCP: List Servers"
- You should see cortex-code listed with a green status

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GETTING STARTED WITH CORTEX CODE

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Use CoCo via Copilot Chat in Agent Mode

HOW TO USE COCO IN COPILOT

- VS Code Copilot Chat has three modes: Ask, Edit, and Agent.
- CoCo tools are available in **Agent mode**.

EXAMPLE PROMPTS

- What tables are in the COCO_WORKSHOP.PIPELINE_LAB schema?
- Write a query to find the top 10 vendors by invoice amount
- Create a Snowflake Task that runs hourly and merges data from my stream into silver
- Explain why this query is slow and optimize it

TIPS

- CoCo respects your Snowflake RBAC -- it can only see/do what your role allows
- You can **reference local files** with @ notation (e.g., **@src/pipeline.sql**)

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Essential Shortcuts & Commands

KEYBOARD SHORTCUTS

Shortcut	Action
Enter	Submit message
Ctrl+J	Insert newline
Ctrl+C	Cancel/Exit
Shift+Tab	Cycle modes (Confirm/Plan/Bypass)
Ctrl+P	Toggle Plan mode
Ctrl+D	Open full screen task view
Ctrl+B	Background bash process
Ctrl+O	Cycle display mode
Ctrl+V	Paste (supports images)

CONTEXT SHORTCUTS

Shortcut	Action	Example
@	File reference	@src/app.py
@file$10-50	Specific lines	@config.yaml$1-20
$	Invoke skill	$semantic-view-optimization
#	Snowflake table	#DB.SCHEMA.TABLE
!	Run bash command	!git status
/	Slash commands	/help, /sql, /plan
?	Quick help	?

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CORTEX CODE WORKSHOP

Data Engineering + Operations

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